

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

M2 Practical-7

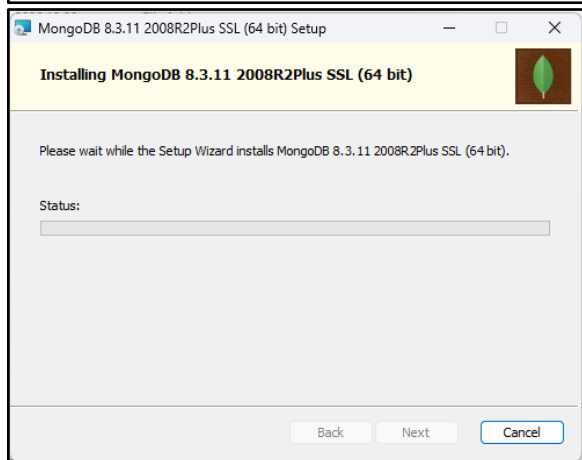
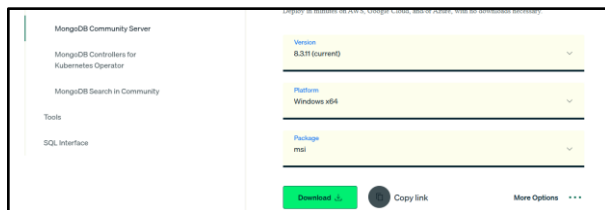
Roll No.:T017	Name: Radhika Mali
Class:TYIT	Batch:1
Date of Assignment: 19-9-2026	Date/Time of Submission: 19-9-2026

Aim: MongoDB Basics.

A. Install and start MongoDB.

Step:

1. Go to the official MongoDB website and download MongoDB Community Server for Windows:
2. [MongoDB Community Server Download](#)
3. Choose:
 - Version: Current/available version
 - Platform: Windows
 - Package: MSI
4. Download the .msi installer.

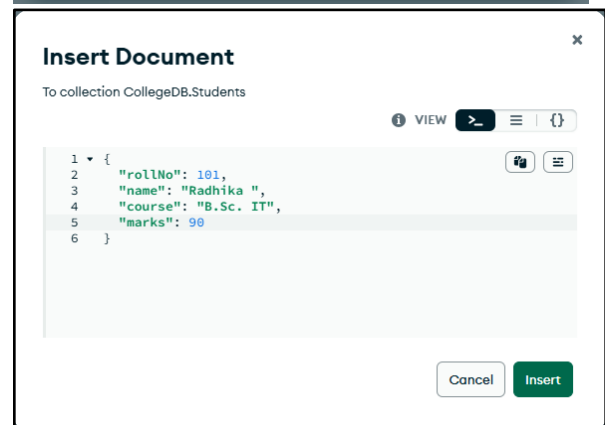
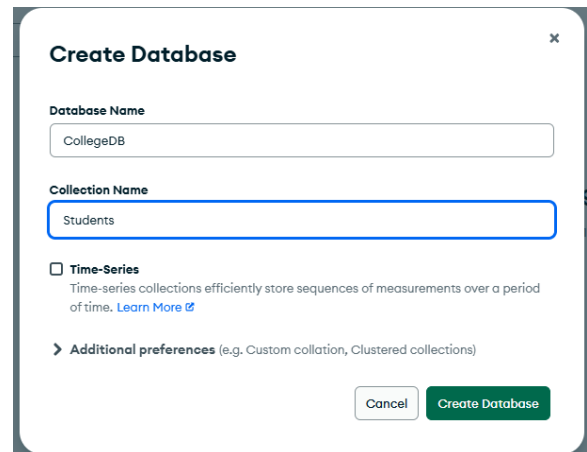


B. Create a database and collection.

Step:

1. Click + symbol beside localhost, it is create database
2. Enter Database Name - CollegeDB , collection name - Students
3. After create database

4. Now click ADD DATA -> Insert document



C. Insert records in MongoDB.

Step:

1. "Open MongoDB Shell" button at the top-right, click: Open MongoDB Shell
2. Command to choose database, CollegeDB prompt will start.

```
> _MONGOSH
> use("CollegeDB")
< switched to db CollegeDB
> db["Students"].find()
< {
  _id: ObjectId('6aae2a073fc644f810ba060d'),
  rollNo: 101,
  name: 'Radhika ',
  course: 'B.Sc. IT',
  marks: 90
}
```

3. insertOne() → inserts one document
4. After enter it will acknowledged

```
> db.students.insertOne({})
< {
  acknowledged: true,
  insertedId: ObjectId('6aae2ad4617e6ccf042f0c84')
}
```

5. insertMany() → inserts multiple documents
6. After enter

```
> db.students.insertMany([{"rollno":102, name : "Abhi", course:"IT"}, {"rollno":103, name : "Aashu", course:"IT"}])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aae2c65617e6ccf042f0c85'),
    '1': ObjectId('6aae2c65617e6ccf042f0c86')
  }
}
```

D. View records from the collection.

Step:

1. db.students.find() - View all records. This displays all student documents.
2. db.students.find().pretty() - View records in a readable format
3. In mongosh window both outputs will be same.

```
db.students.find()
{
  _id: ObjectId('6aae2ad4617e6ccf042f0c84')
}
{
  _id: ObjectId('6aae2c65617e6ccf042f0c85'),
  rollno: 102,
  name: 'Abhi',
  course: 'IT'
}
{
  _id: ObjectId('6aae2c65617e6ccf042f0c86'),
  rollno: 103,
  name: 'Aashu',
  course: 'IT'
}
{
  _id: ObjectId('6aae2e13617e6ccf042f0c87'),
  rollNo: 101,
  name: 'Radha',
  course: 'IT'
}
```

4. Find a particular student - find student with roll number 101:

```
db.students.find({rollNo:101})
{
  _id: ObjectId('6aae2e13617e6ccf042f0c87'),
  rollNo: 101,
  name: 'Radha',
  course: 'IT'
}
```

5. You can find anything from the table.

E. Update and delete records.

Step:

1. Delete - Delete the student with roll number 104.
2. deleteOne - deletes only 1 document
3. deleteMany - deletes all matching documents

```
db.students.deleteOne({rollNo:103})
{
  acknowledged: true,
  deletedCount: 0
}
```

4. Update - Radha's course need to be changed from "IT" to "CS".
5. updateOne - updates only 1 document
6. updateMany - updates all matching documents

```
db.students.updateOne({rollNo:101},{set:{course:"CS"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Output:

```
db.students.find()
{
  _id: ObjectId('6aae2ad4617e6ccf042f0c84')
}
{
  _id: ObjectId('6aae2c65617e6ccf042f0c85'),
  rollno: 102,
  name: 'Abhi',
  course: 'IT'
}
{
  _id: ObjectId('6aae2c65617e6ccf042f0c86'),
  rollno: 103,
  name: 'Aashu',
  course: 'IT'
}
{
  _id: ObjectId('6aae2e13617e6ccf042f0c87'),
  rollNo: 101,
  name: 'Radha',
  course: 'CS'
}
```

F. Connect MongoDB with Node.js.

Steps:

1. Open folder in vs code
2. Run npm init -y to create package.json file.
3. Add server.js file
4. Run npm install mongoose
5. Mongoose is a Node.js library that helps Node.js communicate with
6. MongoDB.
7. Add following code into server.js file

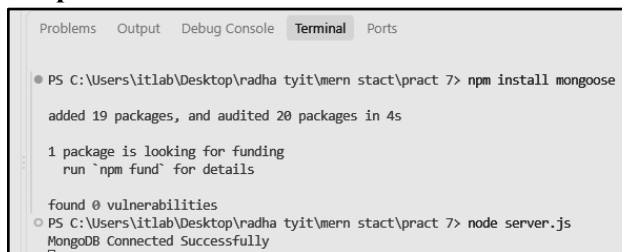
Code:

```
const mongoose = require("mongoose");
```

```
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
```

```
.then(() => {
  console.log("MongoDB Connected Successfully");
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
```

Output:



```
PS C:\Users\itlab\Desktop\radha tyit\mern stack\pract 7> npm install mongoose
added 19 packages, and audited 20 packages in 4s

1 package is looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\itlab\Desktop\radha tyit\mern stack\pract 7> node server.js
MongoDB Connected Successfully
```

8. To view data into terminal

Code:

```
const mongoose = require("mongoose");
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
```

```
.then(async () => {
  console.log("MongoDB Connected Successfully");
```

```
const students = await mongoose.connection.db
  .collection("students")
  .find()
  .toArray();
```

```
console.log(students);
```

```
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
```

Output:



```
PS C:\Users\itlab\Desktop\radha tyit\mern stack\pract 7> node server.js
MongoDB Connected Successfully
[
  { _id: new ObjectId('6aae2ad4617e6ccf042f0c84') },
  {
    _id: new ObjectId('6aae2c65617e6ccf042f0c85'),
    rollno: 102,
    name: 'Abhi',
    course: 'IT'
  },
  {
    _id: new ObjectId('6aae2c65617e6ccf042f0c86'),
    rollno: 103,
    name: 'Aashu',
    course: 'IT'
  },
  {
    _id: new ObjectId('6aae2e13617e6ccf042f0c87'),
    rollno: 101,
    name: 'Radha',
    course: 'CS'
  }
]
```